pfi

# Python for Fixed Income

COURSE 3 · MARKET DATA AND TIME SERIES

# Market Calendars and Cleaning

WEEK 1 · DAY 3

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

# Which days did the market have?

**The market's days are the dates the data has. No holiday list decides them.**

| | |
|---|---|
| `missing_weekdays` | the weekdays a series has no row for |
| `align` | series side by side, no days invented |
| `stale_runs` | a value repeated day after day |

**A holiday is a gap, never a zero change. A missing day is never filled for a calculation.**

### Today's running data

Treasury par yields: 2,940 market days, 2015 to 2026, in percent. Mostly the 10 Yr.

course3_synthetic_spreads.csv: invented for this course, in bp. Not ICE, Moody's, or any index.

# Calendar days, weekdays, market days

**4,292**

calendar days: pd.date_range

**3,066**

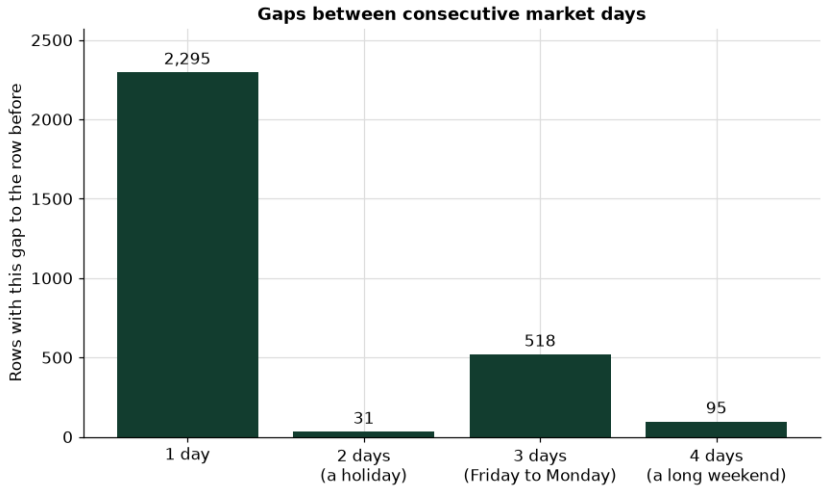
weekdays: pd.bdate_range

**2,940**

market days: the index

**126**

weekdays with no row




**A change is from the previous row, whatever the gap. The Tuesday after a Monday holiday is a change from Friday.**

Mondays have the fewest rows: 538.

# The file as its own calendar

```
def missing_weekdays(index):
    _check_index(index, "index")
    # every Monday to Friday in the span
    weekdays = pd.bdate_range(index[0], index[-1])
    # the ones with no row
    return weekdays.difference(index)
```

### _check_index stops on

an index that is not dates

a date that appears twice

dates not in ascending order

## 126

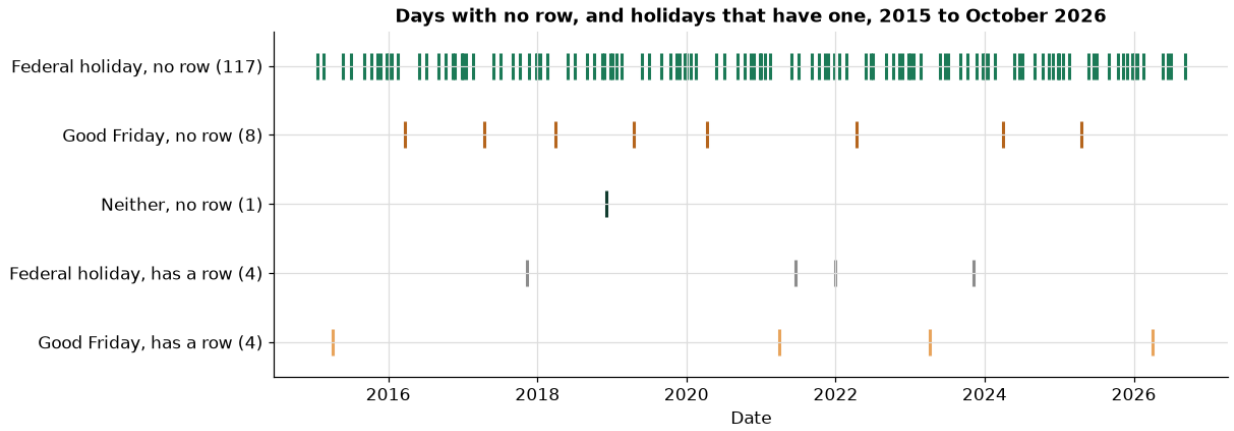
weekdays with no row in the Treasury file

## 2026-09-07

the last one: a Monday

**No holiday calendar is used: the answer is every weekday the data skipped, for whatever reason.**

# A holiday calendar is not the market calendar




## Federal holidays with a row

| | |
|---|---|
| 2017-11-10 | Veterans Day |
| 2021-06-18 | Juneteenth |
| 2021-12-31 | New Year's Day |
| 2023-11-10 | Veterans Day |

**121**
federal holidays

**117**
of them have no row

**8**
Good Fridays, no row

**8**
holidays with a row

**No reason is given for any of these days. A holiday calendar is a guess at the record; the file is the record.**

# FRED's dots and Treasury's absent rows

### Treasury's file

No curve that day: no row at all.

```
2026-09-04, ... 2026-09-08, ...
```

### FRED's API

No value that day: a row whose value is a dot.

```
{"date": "2026-09-07", "value": "."}
```

The FRED-format sample: 43 rows, one NaN, on 2026-09-07, a weekday the file has no row for.

### A live cell, two ways

**No key (the saved notebook):**

```
Live cell skipped: no FRED key was found.
```

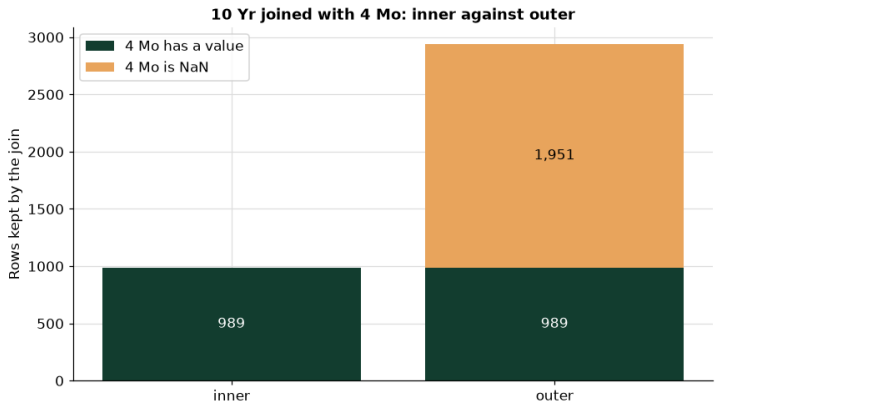
**With your key:**

DGS10 since 2015: every dot is one of the file's missing weekdays

# Joining two series: align

`how="inner"` only the dates every series has

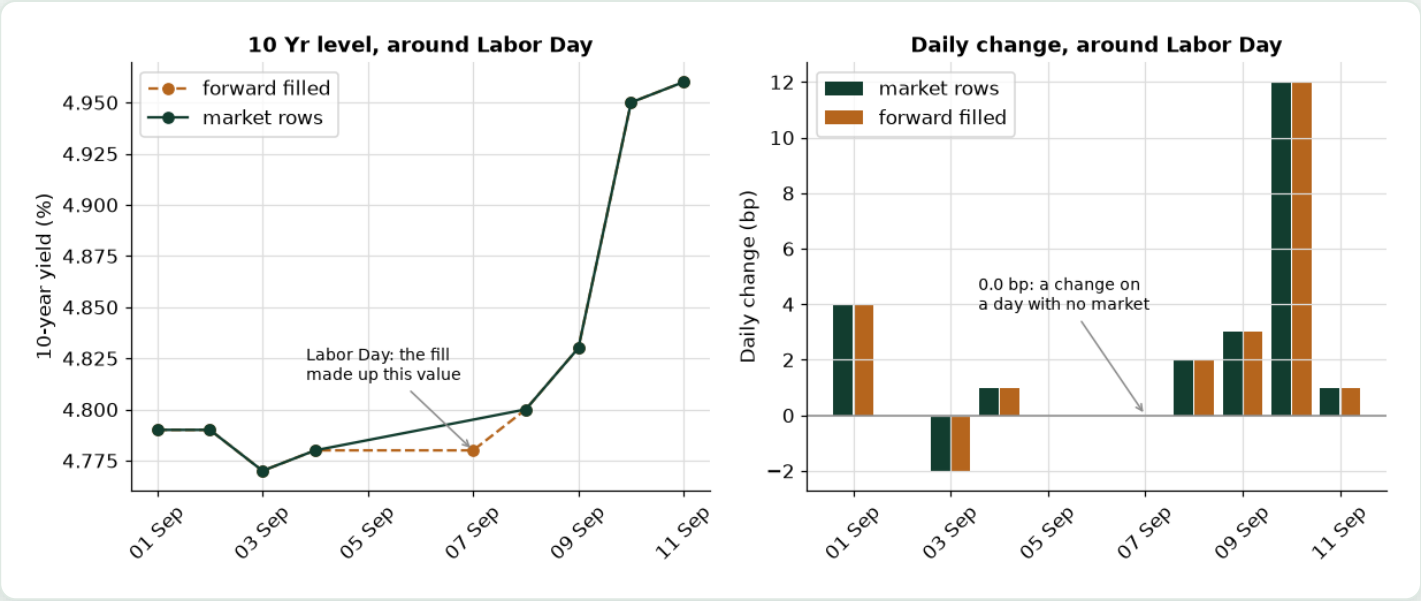
`how="outer"` every date any series has; NaN where a series has no row




| Join | Rows |
|---|---|
| File and FRED sample, outer, dots kept | 43 |
| File and FRED sample, inner, dots dropped | 42 |
| 10 Yr and 4 Mo, inner | 989 |
| 10 Yr and 4 Mo, outer (1,951 NaN) | 2,940 |

**An outer join puts NaN in the gaps, and can bring in a date the market never had. align never fills.**

# Forward fill: for display only

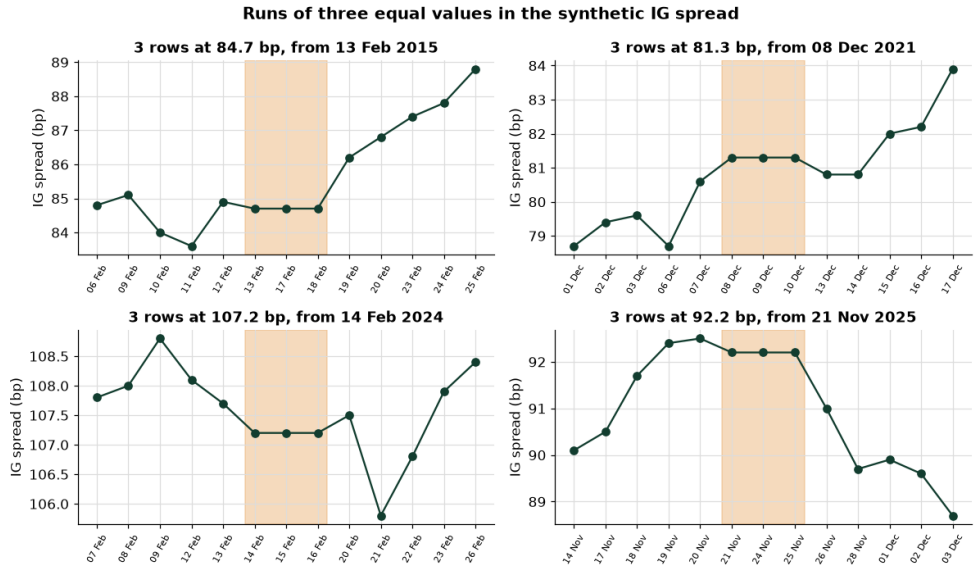



| Over the span | Market | Filled |
|---|---|---|
| Changes | 41 | 42 |
| Zero changes | 3 | 4 |
| Mean absolute, bp | 3.976 | 3.881 |
| Standard deviation, bp | 4.909 | 4.854 |

**ffill put Friday's 10 Yr on Labor Day: a change of 0.0 bp on a day with no market.**

**Never filled for a calculation. Forward fill is for display: on a copy, labelled, never before a change, a window, or a join.**

# Stale values: stale_runs




```
run_id = (series != series.shift()).cumsum()
```

**Four new tests; 13 passed. Table columns: runs of 3 or more, 5 or more, 2 or more rows.**

## A run is a reason to look

Real data repeats: the 2 Yr's longest run is 8 rows. stale_runs says where; another source, or the person who sends the file, says whether.

| Runs | 3+ | 5+ | 2+ |
|---|---|---|---|
| ig_spread_bp (synthetic) | 4 | 0 | 107 |
| hy_spread_bp (synthetic) | 0 | 0 | 29 |
| Treasury 2 Yr | 84 | | |

# Excel side by side

| Excel (tested) | Returns |
|---|---|
| `=WEEKDAY(DATE(2026,9,4),2)` | 5 (Friday) |
| `=WEEKDAY(DATE(2026,9,6))` | 1: Sunday is 1 |
| `=NETWORKDAYS(A2,A2941)-COUNT(A2:A2941)` | 126 |
| `=NETWORKDAYS(DATE(2026,1,1),DATE(2026,10,2))` | 197 |
| `=NETWORKDAYS(DATE(2026,1,1),DATE(2026,10,2),H2:H8)` | 190 = 2026 rows |
| `=WORKDAY(DATE(2026,9,4),1)` | 2026-09-07 |
| `=WORKDAY(DATE(2026,9,4),1,H2:H8)` | 2026-09-08 |
| `=XLOOKUP(DATE(2026,9,7),A2:A2941,B2:B2941)` | #N/A |
| `=AND(C4=C3,C3=C2), filled; COUNTIF TRUE` | 4 |

Tested in Excel 16.0 build 20430, 2026-10-04. H2:H8 holds the missing weekdays of 2026.

# Save the day: amend a message

```
git commit -q -m "Add missing_weekdays, align, and stale_run"
git commit --amend -q -m "Add missing_weekdays, align, and stale_runs, with four tests"
git log --oneline     # one commit, a new short hash
```

### 1. Spot it

stale_run: the function is stale_runs, and the tests are not named.

### 2. Amend

--amend replaces the last commit: same files, new message, new hash.

### 3. Only yours

Amend only what nobody else has. Pushed? Fix with a new commit.

# The AI check

The docstring is the prompt. The code that comes back runs, and has one data bug.

1

## Docstring

Units, conventions, and what it must never do: what you asked for.

2

## Code back

Written for the course in the style of an AI answer. Read it.

3

## Test first

Assert what the docstring promises. Run it: it fails.

4

## Fix

Find the rule it broke, fix it, run the test again.

**Today: closed_days builds the closed days from the federal holiday calendar: 121 dates, not 126, and no Good Friday.**

*Code that runs is not the same as code that is right.*

# What you can do now

- Count calendar days, weekdays, and market days; find missing weekdays
- Say why a holiday calendar is not the market calendar
- Join series with align, inner or outer, and never fill a gap
- Keep forward fill for display; find stale runs, and look
- Count market days in Excel with a holiday list from the data

**Week 1, Day 4** Resampling and the as-of value: which observation stands for a week or a month